

Dayna Blackwell

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Backend Developer & Cloud Engineer with 5 years of experience building data pipelines, event-driven systems, and APIs (REST, GraphQL, JSON-RPC) on AWS. Author of "You Don't Know JSON" (107,000 words on JSON architecture, streaming, and production patterns). Skilled in scalable data ingestion workflows, performance optimization, and infrastructure-as-code (CDK, Terraform). 3x AWS Certified.

PROFESSIONAL SKILLS

- **Programming:** Java, Python, Go, Bash, TypeScript, JavaScript / Node.JS, Rust
- **Architecture Patterns:** REST APIs, gRPC/Protocol Buffers, Event-Driven (SNS/SQS/EventBridge), Pub-Sub, Message Queues, Distributed Caching
- **Backend Frameworks:** Java EE Core, Spring/SpringBoot, FastAPI, Flask, Axum, Express.js
- **Databases:** Oracle SQL, MySQL, PostgreSQL, NoSQL / MongoDB / DocumentDB, Redshift, DynamoDB
- **DevOps / Automation:** Bash scripting, Terraform, CDK, Chef, Jenkins, Github Actions
- **AWS Cloud:** EC2, S3, ALB/NLB, API Gateway, VPC, Lambda functions, EventBridge, SNS, SQS, Step Functions, CloudWatch, Route53, Autoscaling, IAM Policy, Athena, Glue
- **Azure Cloud:** VM, VNet, LB, Entra ID / Azure RBAC, Storage Accounts
- **Monitoring/Observability:** Datadog, Splunk, CloudWatch, Grafana, SolarWinds

PROFESSIONAL EXPERIENCE

Blackwell Systems

Author, Editor & Publisher

<https://leanpub.com/you-dont-know-json>

- Authored and edited "**You Don't Know JSON: Beyond Six Data Types**" (2026) - comprehensive technical book (107,000 words, 838 pages) covering JSON ecosystem architecture and production patterns.
- Topics include: schema validation, binary formats, streaming systems, JWT security, and API design at scale.

Best Western Hotels & Resorts

Backend Enterprise Developer

Remote

July 2022–Present

- Led the **modernization of the legacy promotions system for a major hospitality brand** by designing a **serverless event-driven pipeline** (AWS CDK, Lambda, S3, DynamoDB, SNS), introducing real-time stay event processing and cloud-native architecture patterns.
- Built **stay-adapter Lambda** querying **Redshift** for continuous stay data, **emitting events to SNS**; **EventBridge pipeline** filters and routes to **SQS** for consumption by **Python-based promotion rules engine**; all activity recorded in **DynamoDB** using **single table design**.
- Created **Glue** job to transform **DynamoDB single table design** into 4 relational tables for **Redshift** data warehousing.
- Used **Redis Hashes and Sets** to track user progress toward promotion goals in real-time, allowing rules engine to make millisecond-decisions without hitting the Redshift/Oracle backend.
- Participated in the **refactoring** of a **legacy JSP internal application** into a secure, **cloud-native platform** hosted on **CloudFront** with an **Angular frontend** and **Java-based web services** running on **EC2 instances**; implemented **API Gateway** with a custom **Lambda authorizer** for **Okta SSO integration**, private VPC routing via Transit Gateway, and fully defined infrastructure using **Terraform**.
- Evaluated and migrated caching workloads from **ElastiCache Redis** to **Valkey** on **Graviton3** instances, achieving a 20% reduction in compute costs while improving p99 latency by 15%.
- Responsible for developing services and processes to support '**promotion progress indicator**', an engaging customer facing widget presenting information about **seasonal nationwide** and **major international promotions**.
- Developed a high-integrity data ingestion pipeline using **Python and DynamoDB**, implementing **selective field hashing** to filter out system-generated noise and ensure business-logic idempotency for stay events.
- **Built and maintained Java EE REST APIs** supporting Best Western Loyalty programs.

- **Implemented** Givex **RPC integrations** for gift card issuance to loyalty members as part of promotional reward programs.
- Converted existing **infrastructure configurations** to **CDK, Terraform** and **Chef configurations**.
- Creation of automation jobs and process flows written in **Python**, run by **Enterprise Automation Software: Application Manager** (CA Enterprise Software / BroadCom).
- **PL/SQL Database programming** for Oracle Stored Procedures, triggers, etc.
- **Conversion** of PL/SQL Database logic into Java Web Service layer logic.
- Built **data infrastructure** including **ad hoc SQL** reporting for Marketing teams, database integrations (**Oracle, MongoDB, DocumentDB**), and **ETL pipelines** using **Redshift** and **Glue**.
- Provide **on-call support** for revenue-critical production systems, troubleshooting service degradations and outages using **Splunk, Datadog**, and **CloudWatch** in a 24/7 environment.
- Creation of **technical documentation** to facilitate the better use and understanding of Best Western Loyalty Applications and APIs.
- Frequent contributor and/or project leader on initiatives identified in Technology Management's enterprise-wide "Top 15 projects" list.
- Participate in **frequent cross-functional projects** between Loyalty Applications, Loyalty Automation, Marketing, Marketing Analytics and Brand.com teams.
- **Communicate technical concepts** to non-technical stakeholders and Product Owners.

Best Western Hotels & Resorts
Operations Lead

Hybrid – Phoenix, AZ
February 2022–July 2022

- **Leading** and **organizing remediation efforts** during Priority outage / degradation events.
- Performing **root cause investigation and analysis** for degradations and outages of Tier 1,2 Revenue generating applications
- **Created new tools and dashboards in Datadog** for use by Operations department as well as Product teams.
- Administer enterprise certificate authority / **public key infrastructure**.
- Utilize monitoring, tracing and aggregation tools such as **Splunk, Datadog, Grafana, SolarWinds, VPC Flow Logs, CloudWatch, Akamai**, etc to troubleshoot / analyse application performance and contribute to incident research, root cause analysis and system optimizations.
- **Reliability and performance monitoring** for Tier 1,2 critical revenue generating applications.
- Participated in the **migration of enterprise monitoring system** from Nimsoft to Datadog.
- Participated in **migration activities** for applications moving from **on-premises to AWS**.
- **Ticket Management** using **Cherwell** and **Jira**.

CERTIFICATIONS

AWS: Solutions Architect Associate, Developer Associate, AI Practitioner

Azure: Fundamentals (AZ-900), Data Fundamentals (DP-900)

Other: Terraform Associate (Hashicorp), OCI Foundations (Oracle), PCEP Python (OpenEDG Python Institute)

OSS PROJECTS

domainstack (Rust) | <https://docs.rs/domainstack>

Full-stack validation ecosystem for Rust web services with OpenAPI schema generation, TypeScript/Zod codegen, and WASM browser validation. Workspace of 9 crates published to crates.io with framework adapters for Axum, Actix-web, and Rocket.

blackdot (Go, Zsh, Powershell) | <https://blackwell-systems.github.io/blackdot/>

Cross-platform dotfiles framework with multi-vault secrets, templating, and hooks; integrates with Claude Code, AWS, CDK, Python, Go, Rust, Docker; includes extracted libraries and testing infrastructure.

dotClaude: (Go) | <https://blackwell-systems.github.io/dotclaude/>

Claude Code profile management system; integrates with Blackdot for context-aware AI development workflows.

Vaultmux (Go) | <https://blackwell-systems.github.io/vaultmux/>

Unified secrets API supporting Bitwarden, 1Password, pass, AWS, GCP, Azure, Windows Credential Manager.

Pipeboard (Go, Swift, Kotlin) | <https://blackwell-systems.github.io/pipeboard/>

Cross-platform clipboard routing system with open-source terminal CLI, proprietary iOS/Android apps and RESTful mobile backend.

EDUCATION

Dakota State University

Madison, SD

Associate of Science in Computer Science

Certificate: Object Oriented Programming & Design